

Why Do We Need OSPF?

Before the emergence of OSPF, the Routing Information Protocol (RIP) was the most widely used IGP.

RIP is a distance-vector routing protocol which is gradually being replaced by OSPF, due to the former's slow convergence, tendency to form routing loops, and poor scalability.

OSPF is a link-state routing protocol featuring:

- Multicast packet transmission, reducing load on the routers that do not run OSPF
- Support for Classless Inter-Domain Routing (CIDR)
- Load balancing among equal-cost routes
- Packet authentication

With the preceding advantages, OSPF is widely accepted and used as an IGP.

<https://info.support.huawei.com/info-finder/encyclopedia/en/OSPF.html>

6.2.5 OSPF Message Types

There are five OSPF message types in total, and Fig. 6.6 identifies the message types used in each phase of the OSPF operation.

Type 1: Hello packets, which are used to discover and maintain neighbors.

Type 2: Database Description (DD) packets, which are used to send summary information to neighbors about all link state entries in their own link state database. The summary information is about all routers (Router-IDs) in the link state table.

Type 3: Link State Request (LSR) packets, which are used to request detailed information from the other party about the missing link state information of a router.

Type 4: Link State Update (LSU) packets, which are used to send detailed link state information. Routers use such packets to inform neighboring routers of their link state. In OSPF, only LSU needs to display an acknowledgement.

Type 5: Link State Acknowledgement (LSAck) packets, which are used to acknowledge the LSU.

6.2.6 Network Types Defined by OSPF

Before explaining the designated router (DR) and backup designated router (BDR), it is necessary to understand the OSPF network types.

The OSPF network type is an imperative interface variable that affects the operation of OSPF on the interface, for example, what method is used to send OSPF protocol messages, and whether the designated router and backup designated router need to be elected. The default OSPF network type for an interface depends on the data link layer encapsulation used by the interface.

OSPF defines four network types, namely, point to point (P2P) network, broadcast multi-access (BMA) network, non-broadcast multiple access (NBMA) network, and point to multiple point (P2MP) network.

In general, the network type of the OSPF interfaces at both ends of the link must be the same, otherwise the neighbor relationship cannot be established between the two. The OSPF network type can be manually modified on the interface via commands to suit different network scenarios. For example, a BMA network can be modified to a P2P network.

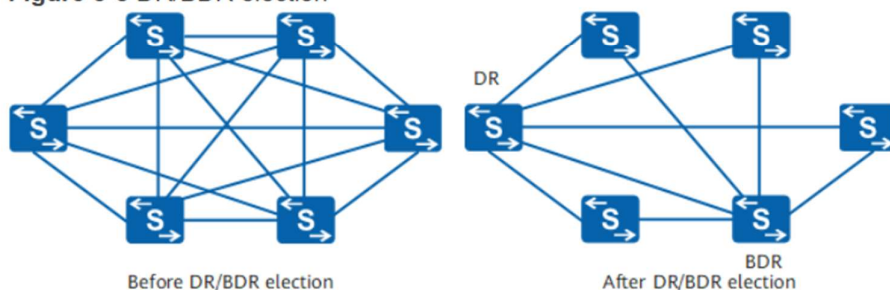
Reason for DR/BDR Election

On broadcast and NBMA networks, any two routers need to exchange routing information. As shown in Figure 5-8, there are n routers that need to establish $n \times (n - 1)/2$ adjacencies. A route change on any router is transmitted to the other routers, which consumes bandwidth resources.

To solve this problem, the concept of DR is defined in OSPF. After a DR is elected, all the other routers send routing information only to the DR, and the DR broadcasts LSAs.

To prevent service interruption caused by DR re-election if the DR fails, a BDR is also elected during DR election. The routers, excluding the DR and BDR, are called DR others. The DR others do not establish adjacencies or exchange any routing information with each other. This reduces the number of adjacencies established between routers on broadcast and NBMA networks.

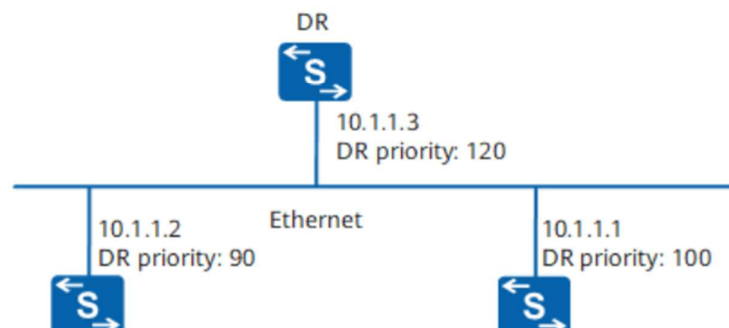
Figure 5-8 DR/BDR election



Election Principle

The DR and BDR are not designated manually, but are elected by all routers on the local network segment. In Figure 5-9, DR priorities of router interfaces determine whether the interfaces are qualified for DR/BDR election. Routers with DR priorities greater than 0 on the local network segment can be considered as candidates. The ballots in this election are performed using Hello packets. Each router adds elected DR information to a Hello packet and sends the packet to other routers on the network segment. When two routers on the same network segment declare that they are the DR, the router with a higher DR priority is elected as the DR. If the two routers have the same DR priority, the router with the greater router ID is elected as the DR. The router whose DR priority is 0 cannot be elected as the DR or BDR.

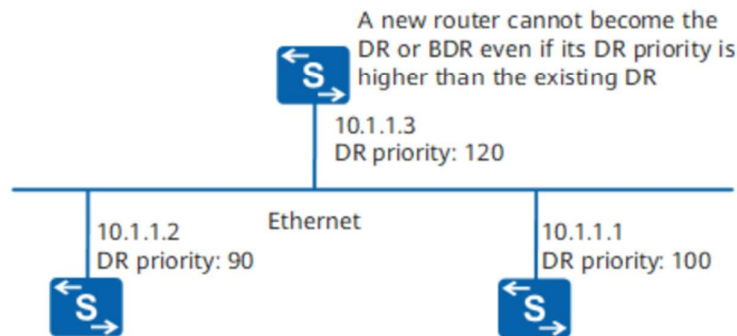
Figure 5-9 Election principle



Non-preemption Principle

A router newly added to a network segment does not initiate a new election. Instead, it checks whether a DR exists on the network segment. In **Figure 5-10**, a DR exists on the network segment. Even if the DR priority of the newly added router is higher than that of the existing DR, the newly added router does not declare itself the DR, and acknowledges the existing DR. Routers on a network segment establish adjacencies only with the DR and BDR. If the DR changes frequently, all routers on the network segment need to establish adjacencies with the new DR and BDR accordingly. As a result, a large number of OSPF packets are transmitted on the network segment within a short period of time, consuming bandwidth resources. The non-preemption principle improves network stability and conserves bandwidth resources. On a broadcast or NBMA network, the two routers that are qualified for DR election and start first become the DR and BDR respectively.

Figure 5-10 Non-preemption principle



The default OSPF priority for a router interface is **1**. This value is used during the Designated Router (DR) and Backup Designated Router (BDR) election process on multi-access networks (like Ethernet). The priority can range from 0 to 255, where a higher value is preferred. data center Cisco Learning Network +3

BDR Cannot Be Elected on the Network

If the **ospf dr-priority** command is run on GE0/0/1 of R2, R3, and R4 to set the interface DR priority to 0, the three routers are unqualified for DR/BDR election and can only work as DR others. Only one router (R1) on the network is qualified for DR/BDR election.

In conclusion, if no router on a broadcast network is qualified for DR/BDR election, the network has no DR or BDR, and routers on the network do not establish adjacencies. In this case, the states of all neighbor relationships among these routers are 2-Way.

<https://support.huawei.com/enterprise/en/doc/EDOC1000178171/32a400ec/fundamentals-of-ospf>

Router ID

A router requires a router ID if it is to run OSPF. A router ID is a 32-bit unsigned integer, uniquely identifying a router in an AS.

A router ID can be manually configured or automatically selected by a router:

- It is recommended that you manually configure a router ID to ensure OSPF stability.
- If no router ID is manually configured for a router, the router selects an interface IP address as its router ID. A router selects its router ID based on the following rules:
 1. The router preferentially selects the largest IP address among loopback interface addresses as its router ID.
 2. If no loopback interface is configured, the router selects the largest IP address among other interface addresses as its router ID.

After the router has been running OSPF and selected its router ID, it still uses this router ID if the interface whose IP address is used as the router ID is Down or disappears (for example, the **undo interface loopback** loopback-number command is run) or a larger interface IP address exists. The router can obtain a new router ID only after a router ID is reconfigured for the router or an OSPF router ID is reconfigured and the OSPF process restarts.

<https://info.support.huawei.com/info-finder/encyclopedia/en/OSPF.html>

Table 21-3 Neighbor Requirements for OSPF

Requirement	Required for OSPF	Neighbor Missing if Incorrect
Interfaces must be in an up/up state.	Yes	Yes
Access control lists (ACL) must not filter routing protocol messages.	Yes	Yes
Interfaces must be in the same subnet.	Yes	Yes
They must pass routing protocol neighbor authentication (if configured).	Yes	Yes
Hello and hold/dead timers must match.	Yes	Yes
Router IDs (RID) must be unique.	Yes	Yes
They must be in the same area.	Yes	Yes
OSPF process must not be shut down.	Yes	Yes
Neighboring interfaces must use same MTU setting.	Yes	No
Neighboring interfaces must use same OSPF network type.	Yes	No

In OSPF (Open Shortest Path First), the RouterDeadInterval is the time a router waits without receiving a Hello packet before declaring a neighbor down.

Default on Broadcast Networks

- HelloInterval: 10 seconds
- RouterDeadInterval: 40 seconds

✦ The RouterDeadInterval is typically $4 \times \text{HelloInterval}$.

OSPF multicast addresses

This simple design gives us a great backdrop from which to observe the results of the broadcast network type on each router. Both interfaces (G0/0 and G0/1) on each router use the broadcast network type and perform the following actions:

- Attempt to discover neighbors by sending OSPF Hellos to the 224.0.0.5 multicast address (an address reserved for sending packets to all OSPF routers in the subnet)
- Attempt to elect a DR and BDR on each subnet
- On the interface with no other routers on the subnet (G0/1), become the DR
- On the interface with three other routers on the subnet (G0/0), be either DR, BDR, or a DROther router
- When sending OSPF messages to the DR or BDR, send the messages to the all-OSPF-DRs multicast address 224.0.0.6

Different states of OSPF

